EVERY FINITE SET OF ELEMENTS OF A BOUNDED ABOVE COMPLEX OF FREE MODULES LIES IN A BOUNDED SUBCOMPLEX OF FINITE FREE MODULES

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Editorial note. The lead-in sentence at source lines 38548-38550, announcing that a simple lemma is needed before the dual and invertible objects of D(R) can be analysed, is outside the counted spans because it belongs to the surrounding discussion. The lemma that follows this one, at source lines 38580-38591, is a different statement and is not counted.

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Allowed standard prerequisites (editorial).

- An element of a direct sum of free modules on one generator each involves only finitely many summands, so the elements of a finite list lie in the span of a finite subset of a chosen basis. used at proof lines 38567-38571, where the finite subsets J' of J and K' of K are chosen. Produces the finite rank pieces out of which the bounded subcomplex is built.
- Descending induction on the lowest degree of the given elements, with the zero complex as the base case when the complex vanishes above that degree. used at proof lines 38562-38566 and again at proof lines 38572-38575. Reduces the statement for a finite list of elements to the statement for the elements of strictly higher degree after the degree a piece has been chosen.
- A subcomplex may be specified by giving its terms in degrees above a and one more term in degree a, provided the differential of that last term lands in the next term. used at proof lines 38571-38577, where the differential of the basis elements b_j is required to land in the span of the chosen c_k. Justifies taking G equal to H in degrees above a and to the span of the b_j

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in degree a, so that G is a subcomplex with finite free terms and bounded support.

Lemma 0.2. *Let R be a ring. Let $F^\bullet$ be a bounded above complex of free R-modules. Given pairs (n_i, f_i), $i = 1, \ldots, N$ with $n_i \in \mathbf{Z}$ and $f_i \in F^{n_i}$ there exists a subcomplex $G^\bullet \subset F^\bullet$ containing all f_i which is bounded and consists of finite free R-modules.*

Proof. By descending induction on $a = \min(n_i; i = 1, \ldots, N)$. If $F^n = 0$ for $n \geq a$, then the result is true with $G^\bullet$ equal to the zero complex. In general, after renumbering we may assume there exists an $1 \leq r \leq N$ such that $n_1 = \ldots = n_r = a$ and $n_i > a$ for $i > r$. Choose a basis $b_j, j \in J$ for F^a. We can choose a finite subset $J' \subset J$ such that $f_i \in \bigoplus_{j \in J'} Rb_j$ for $i = 1, \ldots, r$. Choose a basis $c_k, k \in K$ for F^{a+1}. We can choose a finite subset $K' \subset K$ such that $\mathrm{d}^a_F(b_j) \in \bigoplus_{k \in K'} Rc_k$ for $j \in J'$. Then we can apply the induction hypothesis to find a subcomplex $H^\bullet \subset F^\bullet$ containing $c_k \in F^{a+1}$ for $k \in K'$ and $f_i \in F^{n_i}$ for $i > r$. Take $G^\bullet$ equal to $H^\bullet$ in degrees $> a$ and equal to $\bigoplus_{j \in J'} Rb_j$ in degree a. $\square$